

Inflation Inequality Across the Euro Area

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Household Inflation Exposure in the Euro Area

- How do relative-price shocks affect households with different consumption baskets?
- We construct monthly inflation indices by income, age, and residence for 20 euro-area countries. The indices exclude rents and are calibrated to HICP weights over matched products.
- Counterfactual price paths quantify how fiscal measures changed average inflation and income-group gaps.
- Household microdata reveal the dispersion of individual inflation within groups.

Heterogeneous Effects of Inflation

Distributive effects of inflation operate through three channels.

Channel	Mechanism	Covered here?
Nominal income	The degree to which income is protected against inflation varies across households (the “inflation tax”)	No
Balance sheets	Wealth is transferred from creditors (lenders and savers) to debtors (borrowers) as the real value of debt declines	No
Consumption baskets	Different expenditure shares imply different inflation exposure	Yes

An introductory example

Different baskets generate an inflation gap only when price changes differ across products.

		Inflation of the first income quintile	Inflation of the fifth income quintile	Inflation gap (Q1–Q5)
Expenditure share	Energy	10%	6%	–
	Other consumption	90%	94%	–
Shock 1	Energy prices +40% Other prices unchanged	4.0%	2.4%	1.6 pp
Shock 2	All prices +40% Energy and other consumption	40.0%	40.0%	0.0 pp

What Do Our Inflation Indices Capture?

Our indices do not capture differences in product choices or prices paid within a COICOP category, which require more detailed purchase data.

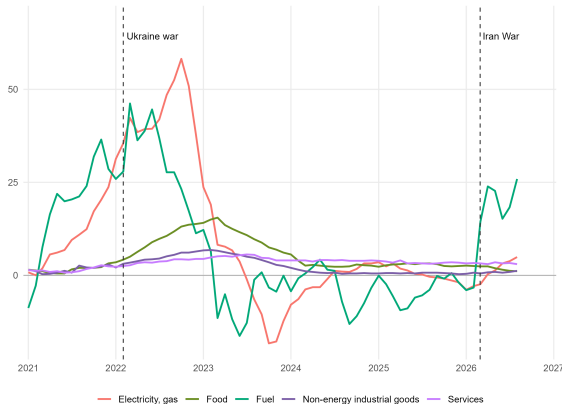
Source of heterogeneity	Example	Our indices
Across categories	Different shares of food, energy and services	Captured
Within categories	Different brands, pack sizes or organic vs. regular products	Not observed
Within identical products	Different retailers, promotions or purchase dates	Not observed

Main Findings, 2021–2023

- The annual Q1–Q5 inflation gap reached **1.2 percentage points in 2022**, with large cross-country differences.
- Price policies lowered inflation by **1.5 percentage points in 2022** and reduced the income gap by **0.6 percentage point**.
- Household-level inflation displays substantial dispersion within income quintiles.
- In 2022, the inflation burden reached **8.7% of net income for Q1**, compared with **4.7% for Q5**.

Energy and Food Drove Large Relative-Price Changes

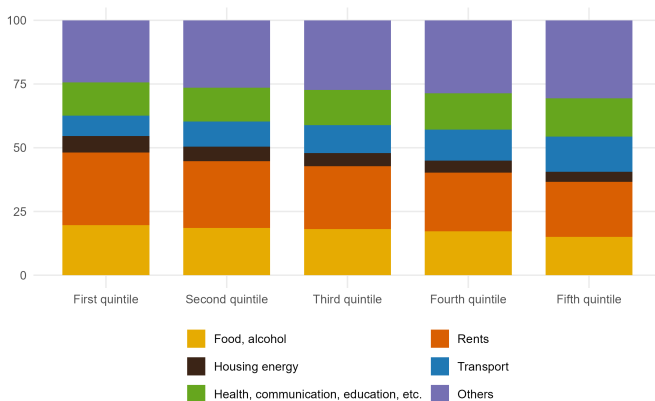
Euro area, year-on-year inflation (%).



Source: Eurostat HICP and HBS; authors' calculations.

Consumption Baskets Differ Across Income Quintiles

Shares of total expenditure including rents, 2020 (%).



Source: Eurostat HICP and HBS; authors' calculations.

Linking HICP Prices to Household Budget Surveys

- **Prices:** monthly HICP components and annual official product weights.
- **Consumption Baskets:** fixed 2020 HBS household expenditure; group baskets are built from microdata where available.
- **Coverage:** 20 countries for group indices; 18 for household distributions, excluding Italy and Portugal.
- **Matching:** remove actual and imputed rents, then calibrate matched three-digit COICOP products.

Calibrated Baskets Produce Chained Inflation Indices

Inflation inequality R package: download, calculate, visualize

github.com/JeremiMontornes/inflationinequality

Calibrate expenditure weights

RAS adjusts the product-by-group matrix M to match group expenditure shares $s_{g,y}$ and HICP product shares:

$$\sum_j M_{j,g,y} = s_{g,y}, \quad \sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}.$$

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}}, \quad \sum_g s_{g,y} \omega_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}.$$

Use expenditure shares, not population shares;
matched products exclude rents.

Construct and chain price indices

Compare product prices with the preceding
December:

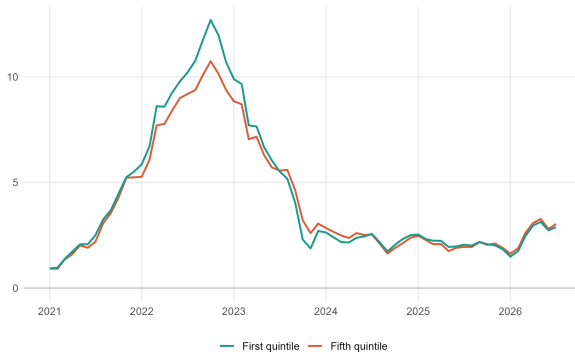
$$P_{g,y,m} = \sum_j \omega_{j,g,y} \frac{p_{j,y,m}}{p_{j,y-1,12}}.$$

$$I_{g,y,m} = I_{g,y-1,12} P_{g,y,m}.$$

Euro-area indices combine country series using
annual HICP country weights.

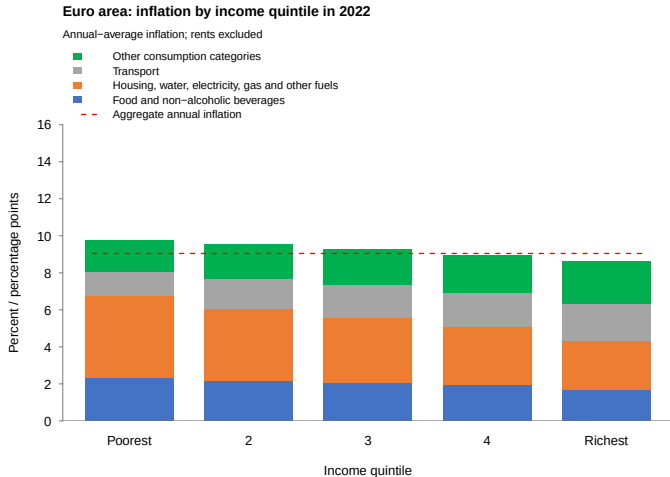
The Income Inflation Gap Peaked During the Energy Shock

ECB analysis reports similar findings using a slightly different methodology (Bobasu et al., 2023).



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Euro Area: Inflation by Income Quintile in 2022

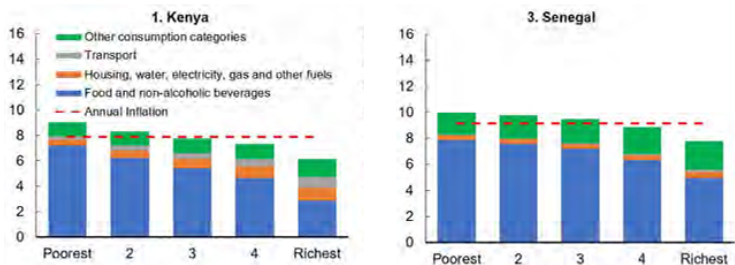


Source: Eurostat HICP and HBS; authors' calculations. Layout: IMF Fiscal Monitor (April 2023), Fig. 2.5.

Euro-area Q1–Q5 gap = 1.2 pp in 2022.

Focus: Kenya and Senegal

Kenya and Senegal, 2021Q2–2022Q2; inflation by quintile (%).



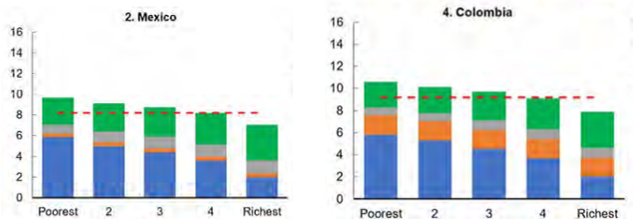
Q1–Q5 inflation gap: Kenya ≈ 2.9 pp; Senegal ≈ 2.1 pp.

- Faster food-price increases weigh more on poorer households because food absorbs a larger share of their spending.

Source: IMF, Fiscal Monitor, 2023, .

Focus: Mexico and Colombia

Mexico and Colombia, 2021Q2–2022Q2; inflation by quintile (%).



Q1–Q5 inflation gap: Mexico ≈ 2.6 pp; Colombia ≈ 2.8 pp.

- Food also drives the Q1–Q5 gap in these countries.

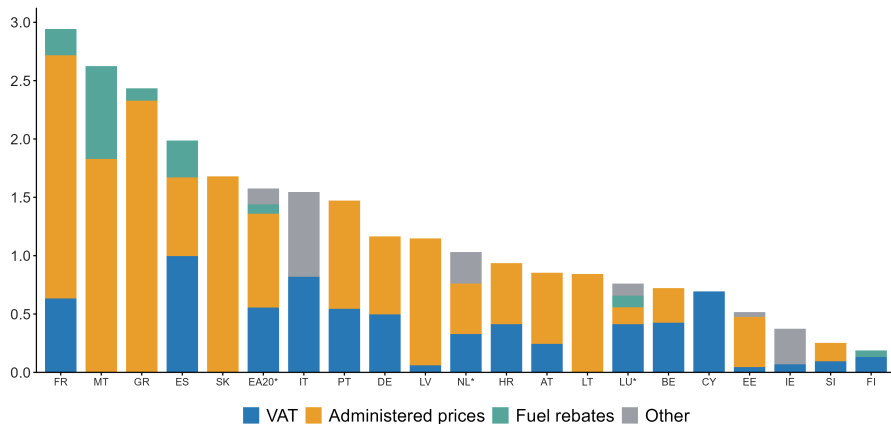
Euro-area Q1–Q5 gap = **1.2 pp in 2022.**

Source: IMF, Fiscal Monitor, 2023

Is there a role for fiscal policy in fighting inflation?

- **Monetary policy:** In response to a supply shock such as that of 2021–2023, central banks tighten policy to keep inflation expectations anchored and prevent second-round effects.
- **Conventional fiscal policy:** The government should tighten the budget to cool demand and, ultimately, inflation.
- **Unconventional fiscal policy:** The government can support households and businesses by reducing inflation directly and can limit its heterogeneous effects through measures targeted at low-income groups.
- **Unconventional fiscal policies can have immediate effects**, while the **transmission lag for monetary policy** is generally estimated to be around 18 months.

Fiscal Costs of Energy Measures (Percent of 2022 GDP)



All 191 active measures have a documented cost; the EA20 total is 1.58% of 2022 GDP. VAT includes excises; “Other” includes envelopes without an instrument split; * indicates an unidentified split.

Source: Bruegel, national sources, Eurostat; authors’ calculations.

France: Defining the Energy-Price Counterfactuals

Gas and electricity: use the CRE tariff paths with and without the shield.

$$\tilde{p}_{j,t} = p_{j,t} \frac{T_{j,t}^{\text{ref}}}{T_{j,t}^{\text{obs}}}.$$

Motor fuels: add the statutory rebate to the observed pump price.

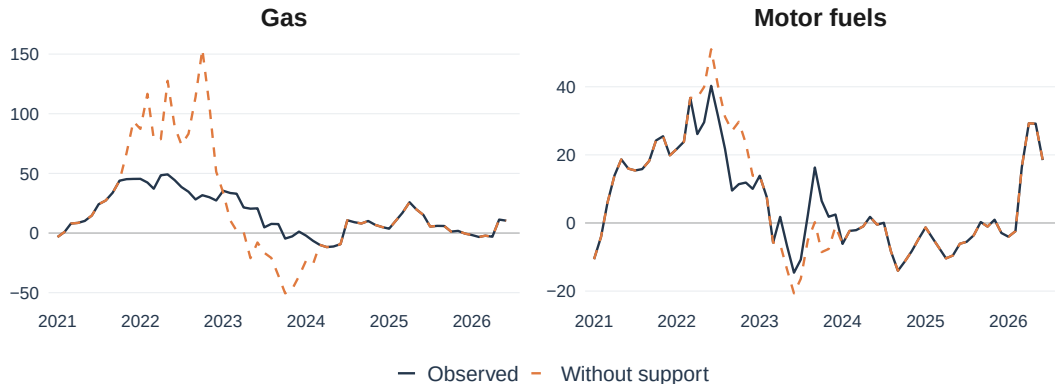
$$\tilde{p}_{\text{fuel},t} = p_{\text{fuel},t} \frac{r_t^{\text{M}} + s_t}{r_t^{\text{M}}}.$$

p : HICP component index; T : regulated tariff; r^{M} : pump price; s : rebate, in euros per litre.

These direct monthly reference paths require no fitted pass-through coefficient.

France: Price Measures Reduced Gas and Fuel Inflation

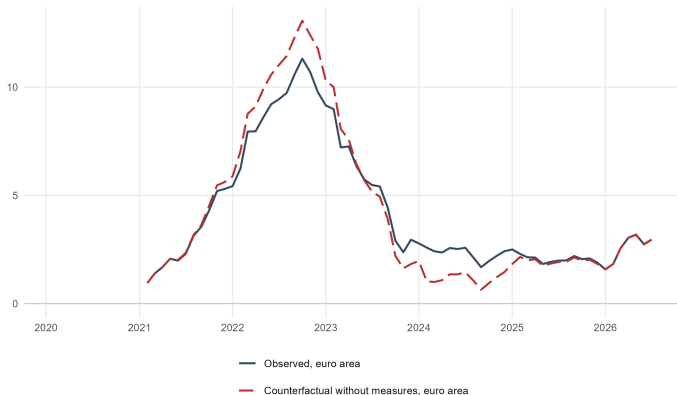
Observed and no-policy year-on-year inflation (%).



Source: Eurostat HICP; authors' counterfactual calculations.

Price Policies Compressed the Inflation Peak

Observed and no-policy year-on-year inflation (%).



Source: Eurostat HICP and HBS; authors' calculations.

Policy Effects on the Q1–Q5 Gap

Euro area (pp)	2021	2022	2023
Observed Q1–Q5 gap	0.1	1.2	0.0
Policy effect on the gap	-0.1	-0.6	-0.5

Price measures narrowed the gap by **0.6 point in 2022** and **0.5 point in 2023**.

Electricity and gas support account for most of this reduction; fuel support partly offsets it in 2022.

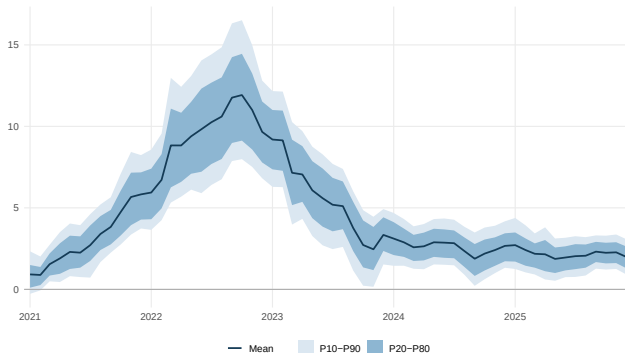
Source: Eurostat HICP and HBS; policy effect = observed minus counterfactual gap.

Household Microdata Show Unequal Inflation Exposure

- **Household-specific baskets:** 152,613 households across 18 countries (excluding Italy and Portugal).
- **Household-specific inflation:** non-rent expenditure shares are matched to country-level HICP price changes.
- **What the measure captures:** inflation differences arising from households' consumption baskets. Common HICP prices do not capture differences in prices paid for the same goods, a major source of household inflation dispersion in Kaplan and Schulhofer-Wohl (2017).

The Inflation Surge Widened Household Dispersion

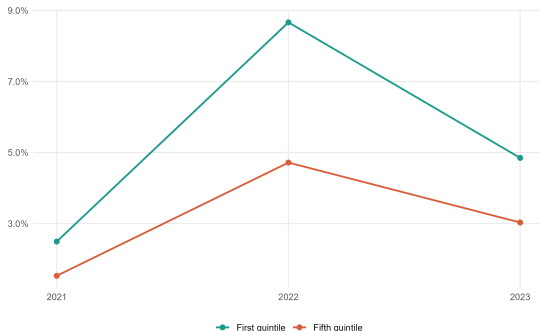
Inflation inequality is larger at the individual level.



October 2022: mean **11.9%**; weighted national P10-P90: **8.0–16.5% (8.5 pp)**.

Source: Eurostat HICP, HBS. National survey-weighted statistics, then annual HICP country weights (18 countries). Bands average national percentiles.

Inflation Burden Is Larger for Low-Income Households



- The Q1 burden reached **2.5%, 8.7%, and 4.8%** of net income in 2021–2023, versus **1.5%, 4.7%, and 3.0%** for Q5.

This is price exposure scaled by 2020 resources, not an observed loss of real income or welfare.

Source: Eurostat HICP and HBS microdata; authors' calculations. 18-country sample; scaled non-rent exposure, in percent of net income. Ratios winsorized at the national 99th percentile.

Conclusion

- **Disaggregated inflation indicators.** HICP prices and household baskets reveal heterogeneous inflation exposure across and within groups.
- **Inflation burden.** In 2022, inflation exposure reached 8.7% of income for Q1, versus 4.7% for Q5.
- **Unconventional fiscal policy.** UFP reduced euro-area inflation and compressed inflation inequality. The counterfactual identifies direct consumer-price effects; production-network effects are analyzed separately for import-price shocks.
- **Effectiveness was state-dependent.** UFP was effective, but monetary policy and favorable circumstances also played a role: the energy-price shock was temporary, and the euro-area economy was not especially overheated, unlike the U.S. economy.

Appendix

Household Energy and Food Widened the Income Gap

Q1–Q5 contribution (pp)	2021	2022	2023
Food and non-alcoholic beverages	0.1	0.7	0.7
Alcohol, tobacco and narcotics	0.1	0.1	0.2
Clothing and footwear	0.0	0.0	-0.1
Housing maintenance and furnishings	-0.1	-0.3	-0.2
Housing energy	0.4	1.9	0.1
Vehicle operation	-0.3	-0.4	0.0
Transport (excluding vehicle operation)	-0.1	-0.3	-0.2
Other	-0.1	-0.5	-0.6
Total	0.1	1.2	0.0

Household Energy Accounts for Most of the Gap Reduction

Product	Inflation effect			Q1–Q5 gap effect		
	2021	2022	2023	2021	2022	2023
Food	0.0	0.0	-0.1	0.0	0.0	0.0
Housing energy	-0.1	-1.2	-0.8	-0.1	-0.6	-0.5
Vehicle operation	0.0	-0.4	0.3	0.0	0.1	0.0
Other products	0.0	0.0	0.0	0.0	0.0	0.0
Total	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5

Household energy lowers the gap by **0.6 point in 2022** and **0.5 point in 2023**; vehicle operation offsets **0.1 point** in 2022.

Policy Effects Differed Sharply Across Countries and Years

Country	Inflation effect			Q1–Q5 gap effect		
	2021	2022	2023	2021	2022	2023
Germany	0.0	-0.7	-1.5	0.0	-0.3	-1.2
France	-0.1	-2.7	-0.6	0.0	-1.1	-0.5
Italy	-0.2	-1.4	0.7	-0.1	-0.2	-0.1
Spain	-0.5	-2.1	-0.3	-0.2	-0.8	-0.3
Netherlands	0.0	-2.5	-0.5	0.0	-1.2	-0.3
Portugal	0.0	-1.0	-1.8	0.0	-0.4	-1.1
Slovakia	0.0	0.0	-3.1	0.0	0.0	-1.7
Euro area	-0.1	-1.5	-0.6	-0.1	-0.6	-0.5

A positive later inflation effect can reflect base effects, while price levels remain lower.

Group and Household Indices Both Exclude Rents

- Exclude actual and imputed rents and corresponding HICP rent components before matching and aggregation.
- Reconstruct parent housing categories without rents and normalize weights over matched non-rent products.
- Both group and household indices use RAS calibration to annual HICP product margins, with matched non-rent coverage.

Rent sensitivity and official all-items validation retain their explicitly stated reference coverage.

RAS Calibration and the Micro–Macro Link

a_{ic} : survey weight; E_{ic} : matched non-rent expenditure; $L_{ic,y,m}$: December-relative household price link.

$$L_{c,y,m} = \frac{\sum_i a_{ic} E_{ic} L_{ic,y,m}}{\sum_i a_{ic} E_{ic}}, \quad L_{y,m} = \sum_c q_{c,y} L_{c,y,m}.$$

$$I_{y,m} = I_{y-1,12} L_{y,m}.$$

$q_{c,y}$: annual HICP country weights, adjusted for rents and normalized over included countries.

Aggregate links before chaining. Calculate inflation from the resulting aggregate index.

The country-weighted average of national household means is a separate distributional statistic.

Inflation Burden Scales Exposure by Household Resources

For household i in country c , the raw exposure measure is

$$\mathcal{B}_{ic,y}^{\text{raw}} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}.$$

- $\bar{\pi}$: annual household inflation (%); X : matched non-rent expenditure; Y : net income, both from the 2020 HBS.
- Winsorize expenditure-to-income ratios at the national survey-weighted 99th percentile, then aggregate household burdens.
- Non-rent inflation is scaled by matched non-rent expenditure relative to income: this measures exposure relative to income, rather than an exact welfare loss.

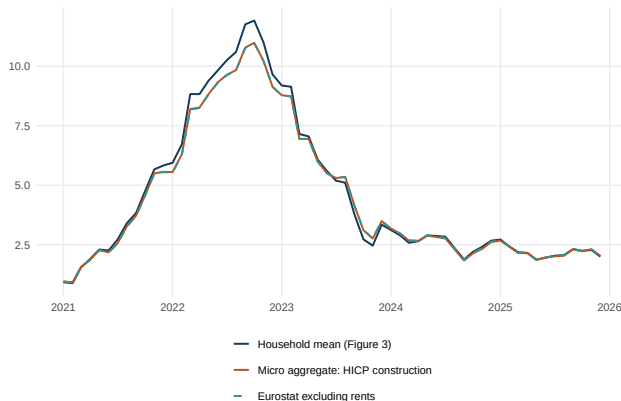
Reconstructing Prices Without the Measures

For product j and month t , multiply the observed HICP index by its counterfactual-to-observed retail-price ratio:

$$\tilde{p}_{j,t} = p_{j,t} \frac{\tilde{r}_{j,t}}{r_{j,t}}.$$

- **Direct reconstruction:** statutory taxes, rebates, or observed reference tariffs, with specified coverage and pass-through.
- **Aggregate-equivalent effects:** fit one parameter so the geometric mean of monthly ratios over the policy period matches the inventory's assumed ratio.
- **Conditional benchmarks:** representative bills or contracts when national counterfactual tariffs are unobserved.

Matching Coverage Clarifies the Micro–Macro Comparison



Finer Product Detail Can Widen or Narrow Income Gaps

2022 Q1–Q5 gap (pp)	Macro D3	Micro D2	Micro D3	Micro D4	Micro D5
Germany	1.4	2.0	1.5	–	–
France	0.2	0.3	0.3	–	–
Spain	1.0	0.8	1.0	–	–
Netherlands	2.8	2.0	3.0	2.5	1.9
Belgium	2.3	2.5	2.3	2.4	2.3
Lithuania	3.4	2.9	3.7	4.9	4.9

Separate digit-depth sensitivity, not the updated baseline. Dutch matched coverage: 99.3% for macro D3, 99.9% for micro D3, and 87.8% for micro D5.

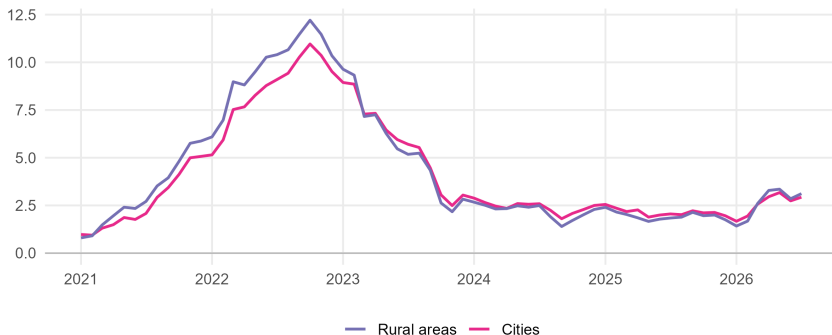
Inflation Inequality Was Highly Uneven Across Countries

Country	2021	2022	2023	Country	2021	2022	2023
Austria	-0.5	0.0	0.5	Ireland	0.5	2.9	1.1
Belgium	1.3	2.3	-2.7	Italy	0.1	1.6	0.5
Croatia	0.0	0.2	-0.3	Latvia	-0.2	4.5	2.5
Cyprus	0.0	0.8	0.2	Lithuania	0.2	3.3	0.4
Estonia	0.6	4.8	1.3	Luxembourg	0.1	0.2	-0.1
Finland	-0.2	-0.9	0.4	Malta	-0.2	0.3	0.6
France	0.4	0.2	0.2	Netherlands	0.0	3.1	-1.2
Germany	-0.4	1.4	0.5	Portugal	0.8	0.2	-0.9
Greece	0.3	1.2	-0.6	Slovakia	-0.9	0.1	0.5
Spain	0.8	0.9	-0.7	Slovenia	0.0	0.4	1.5
Euro area					0.1	1.2	0.0

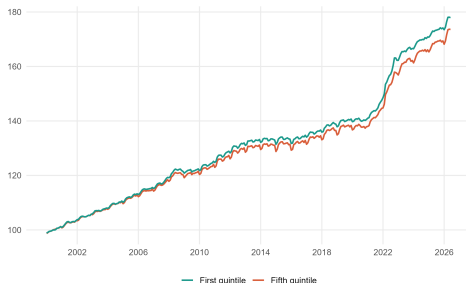
Older Households Faced Higher Inflation



Rural Households Faced Higher Inflation at the Peak



Long-Run Income Gaps Remain Smaller in the Euro Area



The paper's euro-area series contrasts with larger long-run US gaps documented by Jaravel (2024).

National price dynamics can offset one another; broad COICOP categories also miss the within-product price differences captured in US microdata.

Direct and Indirect Exposure to Import-Price Shocks

The observed 2022 extra-EU import-price increases are 83.9% for energy and 21.0% for food.

$$\Delta \mathbf{p}^k = (\mathbf{I} - \mathbf{A}^\top)^{-1} \mathbf{b}^k, \quad \Delta \pi_g^k = \mathbf{w}_g^\top \mathbf{C} \mathbf{d}^k + \mathbf{w}_g^\top \mathbf{C} \Delta \mathbf{p}^k.$$

- $\mathbf{d}^k$: final-import price impulses; $\mathbf{b}^k$: imported-input cost impulses.
- $\mathbf{A}$: FIGARO production network; $\mathbf{C}$: NACE–COICOP mapping; $\mathbf{w}_g$: group expenditure weights.
- Complete pass-through and fixed input coefficients: an accounting exposure simulation, not a causal estimate of observed inflation.

Energy Import Prices Widen the Income Inflation Gap

Product group	Inflation	Q1–Q5 gap	Direct	Indirect
Food and non-alcoholic beverages	0.6	0.2	0.0	0.2
Alcohol, tobacco and narcotics	0.1	0.0	0.0	0.0
Clothing and footwear	0.1	0.0	0.0	0.0
Housing maintenance and furnishings	0.2	-0.1	0.0	-0.1
Housing energy	1.7	1.7	0.5	1.2
Vehicle operation	1.8	-0.5	-0.1	-0.5
Transport (excluding vehicle operation)	1.2	0.0	0.0	0.0
Other	0.8	-0.1	0.0	-0.1
Total	6.5	1.1	0.4	0.7

The energy shock adds **6.5 points** of inflation exposure and **1.1 points** to the gap; housing energy contributes **1.7 points** to the gap.

More Granular Data: Scanner Data and Cheapflation

New datasets are expanding the literature on inflation inequality.

Scanner data: supermarkets, Nielsen, GfK, Kantar.

Advantages	Limitations
Barcode-level prices and quantities	Incomplete coverage: mainly groceries
High-frequency observations	Limited cross-country comparability

- **More household heterogeneity:** individual indices reveal dispersion within income groups and product categories (e.g., organic vs. regular products).
- **UK evidence, 2021Q3–2023Q3:** cheaper varieties had faster price growth; lower-expenditure households bought more of these varieties.
- **A possible mechanism:** thin margins may limit cost absorption. The same absolute price rise implies higher percentage inflation for cheaper goods.

Scanner Data Reveal Dispersion Hidden by Common Prices

United States, Nielsen consumer panel; mean interquartile range (IQR) of annual household inflation.

Laspeyres index construction	Mean IQR (pp)
Broad household baskets, CPI prices	1.61
Barcode baskets, national barcode-average prices	3.99
Barcode baskets, household-level prices	7.33

- Detailed product choices and prices paid reveal additional household dispersion.
- These are averages of date-specific IQRs, not additive variance components.

National Designs Require Different Counterfactuals

- **Germany:** bill rebate on 80% of reference consumption; a representative-bill model anchored on April 2023 prices.
- **Netherlands:** cap on actual consumption up to common limits; the reconstruction removes the discount from CBS bills.
- **Spain and Portugal:** the Iberian mechanism compensates fossil-fuel generators, financed by benefiting buyers and congestion revenues.
- **Other designs:** electricity subsidies in Cyprus, bill credits in Ireland, system-charge reductions and social bonuses in Italy.

German Price Brakes: Design Shapes Incentives

- **Krebs and Weber (2024):** examine conditions under which energy price controls can be optimal.
- **Dertwinkel-Kalt and Wey (2025):** price brakes can induce suppliers to raise prices while reinforcing households' incentives to save energy.

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Question 1: What Generates Inflation Inequality?

Which combination can generate inflation inequality through consumption baskets?

Select one answer.

- A. Different consumption baskets and different price changes across products.
- B. Different consumption baskets when all product prices rise at the same rate.
- C. Differences in wage indexation alone.
- D. Differences in savings and debt alone.

Question 2: What Do Our Inflation Indicators Capture?

Which source of household inflation heterogeneity is captured by our HICP–HBS indicators?

Select one answer.

- A. Differences in expenditure shares across COICOP categories, across groups and households.
- B. Differences in product varieties, brands and pack sizes within the same category.
- C. Differences in retailers, promotions and purchase dates for identical products.

Question 3: Policy Responses to a Supply Shock

Which statements correctly describe the policy response to an adverse supply shock?

Several answers are possible. Select all that apply.

- A. Monetary policy should never respond because inflation originates from a supply shock.
- B. Monetary policy should prevent second-round effects and a de-anchoring of inflation expectations.
- C. Conventional fiscal policy can support the policy mix by tightening aggregate demand.
- D. Temporary, targeted unconventional fiscal measures can support monetary policy, but cannot replace it.